The final exam is cumulative, but will be weighted most strongly towards topics covered after the midterm, e.g. material in problem sets 4-6, which included problems from the following sections of the book:

Chapter 6, Section 8: Poincare Index
Chapter 7, Sections 1-5: gradient systems, Lyapunov functions, Poincare-Bendixson Theorem, Relaxation Oscillators and Excitable systems, e.g. Van der Pol and biased Van der Pol.
Chapter 8, Sections 1-2: Saddle-node, transcritical and Pitchfork bifurcations (of subcritical and supercritical varieties) revisited in 2-dim., Hopf bifurcation (also subcritical and supercritical varieties)
Chapter 9, Sections 1-4: Lorenz equations and chaos, sensitive dependence on initial conditions, Lyapunov exponent, Lorenz map
Chapter 10, Sections 1-4: one-dimensional maps – fixed points, cobwebs, linear stability of fixed-points, period-doubling, logistic map and its orbit diagram, computing stability of n-cycles, period-doubling cascade.

This material emphases bifurcation theory, limit cycles and when you can rule them out and when you can be guaranteed to find them, chaotic dynamics and the striking behavior associated with a one-dimensional map, where we did quite a bit around fixed-points and n-cycles, and their linear stability properties.

Open-ended prompt: what things can happen in nonlinear systems that cannot happen in linear ones? Try to think of the big take-aways around that from the course. What things can happen in one-dimension? What things need to be in at least 2 dimensions? Etc.